

PocketSmart AI

Project Documentation — File 7

1. Project Overview

PocketSmart AI is a GenAI-powered budget and recommendation assistant designed to help users make purchasing and planning decisions across different life needs such as home décor, event planning, and jewelry shopping. The system combines user budget information with AI-powered recommendations to present relevant options in an accessible interface.

2. Project Objectives

- Provide recommendations that take the user's available budget into account.
- Reduce the effort required to search across different products and services.
- Use generative AI to understand user requirements and produce useful recommendations.
- Support multiple recommendation categories and use cases.
- Provide a simple interface for interacting with the recommendation assistant.

3. Functional Documentation

Module	Description
User Input	Collects the user's requirement, category, preferences, and budget.
Budget Analysis	Interprets the available budget and uses it as a recommendation constraint.
AI Recommendation	Generates recommendations based on the user's requirements and budget.
Product/Service Discovery	Supports recommendations across different shopping and planning contexts.
Results Display	Presents generated recommendations to the user through the application interface.

4. AI Processing Workflow

1. User enters a requirement and specifies a budget.
2. The application processes the user's request.
3. The GenAI recommendation layer interprets the requirement and constraints.
4. Relevant recommendations are generated according to the request.
5. The recommendations are presented to the user for evaluation.

5. Technology Stack

Layer	Technology / Component
Frontend	React / Next.js based application interface
Styling	TailwindCSS

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AI	Google Gemini API / Generative AI
Application Logic	Backend services and recommendation logic
Deployment	Web deployment platform / cloud services

6. User Interaction Documentation

The user begins by describing what they want to purchase or plan. The user then provides a budget and any relevant preferences. PocketSmart AI processes the request and generates recommendations that can be reviewed by the user. The interface is intended to keep the decision process simple while exposing the reasoning inputs used by the recommendation system.

7. AI Recommendation Documentation

The recommendation component is designed around the user's stated requirements rather than a fixed product list. The AI receives contextual information such as category, budget, and preferences and generates recommendations accordingly. Because generative AI output can vary, recommendations should be treated as assistance for decision-making rather than guaranteed product facts or professional advice.

8. Project Files and Documentation

- Source code and application configuration files.
- Frontend components and user-interface files.
- Backend or API integration files.
- AI prompt and recommendation logic.
- Environment configuration for API credentials.
- Testing and deployment documentation.

9. Deployment Documentation

For deployment, environment variables containing API credentials should be configured through the deployment platform rather than committed to source control. The production build should be tested after deployment, including the main interface, recommendation request flow, API connectivity, and error handling.

10. Limitations and Future Improvements

- Recommendation quality depends on the information supplied by the user and the AI model.
- Product availability, prices, and external platform information may change.
- Future versions can add stronger product-data integration and real-time price comparison.
- The system can be extended with recommendation history, saved preferences, and user profiles.
- Additional categories and multilingual support can be introduced in future releases.

11. Conclusion

PocketSmart AI demonstrates how generative AI can be applied to budget-aware recommendation and planning scenarios. The project combines a user-friendly application interface with AI processing to turn a user's requirements and budget into structured recommendations. This documentation records the major functional, technical, workflow, deployment, and future development aspects of the project.